Step 1: Install and Import Required NLP Libraries

```
# Install NLP libraries
!pip install nltk spacy

# Import libraries
import nltk
import spacy

# Download NLTK tokenizer
nltk.download('punkt')
```

```
Requirement already satisfied: nltk in /usr/local/lib/python3.12/dist-packages (3.9.1)
Requirement already satisfied: spacy in /usr/local/lib/python3.12/dist-packages (3.8.11)
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Requirement already satisfied: weasel<0.5.0,>=0.4.2 in /usr/local/lib/python3.12/dist-packages (from spacy) (0.4.3)
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Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from jinja2->spacy) (3.0.3)
Requirement already satisfied: wrapt in /usr/local/lib/python3.12/dist-packages (from smart-open<8.0.0,>=5.2.1->weasel<0.5.0,>=0
[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data]   Unzipping tokenizers/punkt.zip.
True
```

Step 2: Load a Short Paragraph

```
text = """Natural Language Processing is a branch of Artificial Intelligence.
It enables machines to understand and process human language."""
```

Step 3: Count the Number of Words in the Text

```
# Download the missing punkt_tab resource
nltk.download('punkt_tab')

# Tokenize text into words
words = nltk.word_tokenize(text)

# Count words
word_count = len(words)

print("Number of words:", word_count)
```

```
[nltk_data] Downloading package punkt_tab to /root/nltk_data...
[nltk_data]   Unzipping tokenizers/punkt_tab.zip.
Number of words: 20
```

Step 4: Convert Text to Lowercase

```
# Convert text to lowercase
lower_text = text.lower()

print("Lowercase text:\n", lower_text)
```

```
Lowercase text:
 natural language processing is a branch of artificial intelligence.
it enables machines to understand and process human language.
```

Start coding or generate with AI.